

# SAMUEL ABOLO

Senior Software Engineer · Backend, ML Platform & Distributed Systems · Python · Go · Cloud

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Software engineer with 5+ years of production experience building backend and platform systems that hold up under real load, and ML infrastructure that ships. Work sits at the intersection of distributed systems, AI platform engineering, and cloud infrastructure. Reduced AI inference latency by 90% and ML deployment time from two weeks to under 30 minutes at TripAdvisor; migrated pipelines from Spark/Kubeflow to Snowflake + Ray, dropping runtimes from ~4 hours to ~40 seconds; and built agentic AI backends on GCP using Vertex AI and Gemini. Strongest in Python (expert), fluent in Go, Java, JavaScript/TypeScript, and C/C++. Operates best where correctness is non-negotiable, failures are expensive, and systems must keep evolving without slowing teams down.

## TECHNICAL SKILLS

|                     |                                                                                                                                                                                                                        |
|---------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Languages           | Python (expert), Go, Java, SQL, JavaScript/TypeScript, C/C++                                                                                                                                                           |
| ML & AI             | PyTorch, TensorFlow, Ray (Serve/Data), Anyscale, Vertex AI, Gemini, Google ADK, Hugging Face Transformers, LangChain, OpenAI, PySpark, RAG, embeddings, transformer architectures, computer vision, NLP, deep learning |
| Backend             | Distributed systems, microservices, event-driven architecture, REST APIs, gRPC, pub/sub messaging, concurrency, FastAPI, Django/DRF                                                                                    |
| Data Infrastructure | PostgreSQL, pgvector, Snowflake, Redis, Kafka, RabbitMQ, MQTT, WebSockets, distributed caching, ETL pipelines                                                                                                          |
| Cloud & MLOps       | AWS, GCP (Vertex AI, Cloud Run), Azure, Kubernetes, Docker, Terraform, ArgoCD, CI/CD (GitHub Actions, GitLab CI, Jenkins), Seldon, Kubeflow                                                                            |
| Observability       | Prometheus, Grafana, Loki, CloudWatch, PagerDuty                                                                                                                                                                       |
| Frontend            | React, Next.js, Vue.js, Tailwind CSS, HTML/CSS                                                                                                                                                                         |
| Foundations         | Data structures & algorithms, probability & statistics, discrete mathematics, software architecture, Linux system administration                                                                                       |

## PROFESSIONAL EXPERIENCE

Senior Software Engineer, AI Platform — **Boostr (via Zazmic Inc)** Nov 2025 – Present · New York, USA (Remote)

- Building an agentic AI backend platform for AdOps automation and media buying on GCP using Vertex AI, Gemini, and Google ADK.
- Implemented a media-buying recommendation agent mapping line items to products, media-plan packages, and object types; presented in a CES product demo.
- Designed distributed embeddings ingestion and indexing pipelines into PostgreSQL with pgvector, enabling semantic retrieval across customer artifacts.
- Built agent orchestration workflows and vector-search infrastructure supporting automated AdOps decisioning at scale; owned end-to-end features across FastAPI services and React internal dashboards.
- Mentored junior developers and shared Python best practices within an agile scrum team.

Software Engineer II, AI Platforms Engineering — **TripAdvisor** Oct 2024 – Nov 2025 · Needham, MA (Remote)

- Reduced AI inference latency by 90%: median execution time dropped from ~15 minutes to ~62 seconds.
- Migrated ML pipelines and the offline feature store from Spark on EKS (Kubeflow) to Snowflake + Ray on Anyscale, cutting pipeline runtime from ~4 hours to ~40 seconds.
- Cut ML deployment time from 2 weeks to under 30 minutes by redesigning core CI/CD and infrastructure workflows; led the rollout of GitHub Actions CI/CD for model-serving under strict monorepo and security constraints, replacing GitLab.
- Architected model-serving infrastructure with Ray Serve and Seldon supporting multiple business verticals; built FastAPI services for internal model-serving and platform tooling.
- Built production observability for ML systems using Prometheus, Grafana, Loki, CloudWatch, and PagerDuty.
- Developed a custom load-testing framework in Java (Java 11 + Spring) to stress-test AI microservices under high concurrency.

Founding Engineer — **Remllo** Jul 2025 – Present · Delaware, USA (Remote)

- Building real-time transaction monitoring and fraud-detection infrastructure for African fintech.
- Architected core backend systems, streaming data pipelines, and rule-based fraud signal detection in production.

Founding Engineer / CTO — **Bookclinic** Dec 2025 – Present · Lagos, Nigeria

- Leading technical architecture and product decisions for a healthcare appointment and booking platform, from system design to feature prioritization.
- Building backend infrastructure, provider-facing tooling, and patient booking flows from the ground up.

Software Engineer (Telematics, Contract) — **Stealth Company** Sep 2025 – Nov 2025 · Lagos, Nigeria

- Built a high-throughput real-time telematics platform using Go, PostgreSQL, Redis, and MQTT/WebSockets for an enterprise fleet-management company.
- Designed ingestion pipelines for high-frequency GPRS, GPS, and vehicle-health data across diverse telematics vendors and device models.
- Engineered distributed microservices using gRPC for low-latency communication across telemetry and analytics components.
- Implemented predictive anomaly detection, maintenance-risk logic, and route-anomaly identification from live streaming sensor data; integrated Prometheus/Grafana monitoring.

AI Engineer (EdTech, Contract) — **ExcelMind** Jul 2024 – Sep 2024 · Lagos, Nigeria

- Built a predictive AI system to forecast WAEC/JAMB exam questions by analyzing historical patterns and topic distributions using neural networks.
- Developed a RAG-powered explanation engine clarifying why each answer choice is correct or incorrect; designed training and evaluation data pipelines.

Software Engineer — **Reconciliation & Finance** — **Credrails** Apr 2024 – Sep 2024 · Nairobi, Kenya (Remote)

- Designed and built financial data APIs and reconciliation backend services using Django and DRF for account aggregation workflows.
- Built a client onboarding platform automating document verification and account setup, reducing onboarding time by ~70%.
- Shipped an LLM-powered transaction-reference extraction feature, improving ledger consistency and reconciliation accuracy.

### Software Engineer — MedTech (Contract) — VG Platform (MyCygnal)

Jan 2024 – Mar 2024 · Austin, TX, USA (Remote)

- Led backend development of a patient-engagement platform, delivering microservices for medication alerts, scheduling, and care insights.
- Implemented event-driven architecture with Kafka for real-time patient notifications; deployed Azure infrastructure (Docker, Kubernetes, ACR) for high availability.

### Senior Software Engineer (Machine Learning) — Quibble — Revenue Management for STR

Nov 2022 – Aug 2023 · San Juan, Puerto Rico (Remote)

- Built an ML-based price recommendation system forecasting optimal rates across a 365-day horizon using TensorFlow.
- Developed computer-vision models for room-type classification (transfer learning on ImageNet) and image-based listing attractiveness scoring.
- Designed ETL pipelines for large-scale property data ingestion and cleaning, increasing dataset reliability by 40%.
- Mentored junior engineers on model deployment, testing, and performance tuning.

### Backend Engineer (Contract) — Prunedge

Feb 2023 – Jul 2023 · Lagos, Nigeria

- Designed and implemented a notification microservice handling push, email, and SMS delivery for thousands of users.
- Built authentication, authorization, and user-invitation systems, and wallet APIs supporting withdrawals, transfers, and deposits in production financial systems.
- Implemented role-based access control (RBAC) to secure agent permissions and sensitive financial operations.

### Software Engineer (Industrial Training) — ProDevs

Jan 2023 – Jun 2023 · Lagos, Nigeria (Remote)

- Developed REST APIs with FastAPI for prescription management and automated patient reminders across medtech and real-estate platforms.
- Built Vue.js frontend interfaces from Figma designs to support full feature delivery.

### Software Engineer (Contract) — Get A Manpower

2023 · Remote

- Inherited and revived an abandoned Laravel/MySQL codebase, stabilizing core platform functionality and launching it to first users.
- Built a React admin portal for complaint management, submission review, and moderation, with analytics and operational reporting views.

### Software Engineer (Contract) — Babcock University — Student Union E-Voting

2021 – 2022 · Ilishan-Remo, Nigeria (On-site)

- Designed and built a secure electronic voting system with a FastAPI backend handling voter registration, authentication, ballot submission, and tallying.
- Implemented RBAC to ensure voting integrity and prevent double voting; built a React frontend for ballot casting, election setup, and real-time results, with focus on auditability under concurrent load.

### Python Developer — Self-Employed

Jan 2018 – Nov 2022 · Remote

- Built and deployed custom software for clients, including inventory systems and trading trackers, plus web apps involving real-time scraping and data analysis.
- Developed an AI-powered medical suggestion API generating personalized health recommendations, deployed to production.

### Software Engineer (Apprenticeship) — Sidmach Technologies Nigeria

Apr 2019 · Lagos, Nigeria

- Built and deployed an ML model to predict student exam performance from practice-test data, serving real-time evaluation on Microsoft Azure.

## EDUCATION

### B.Sc. Software Engineering — Babcock University, Ilishan-Remo, Nigeria

Jan 2021 – Jul 2024

GPA: 4.07/5.0 (3.26/4.0) · Major GPA: 4.28/5.0 (3.42/4.0). Thesis: "Design and Implementation of an ML-Based Predictive Model for Colorectal Cancer Patient Survival" (Grade: A).

Relevant coursework: Distributed Systems, Algorithms & Data Structures, Artificial Intelligence, Big Data Engineering, Database System Design, Software Architecture, Linux System Administration.

### Google Machine Learning Bootcamp 2022 — Sub-Saharan Africa Cohort

Aug 2022 – Dec 2022

Selected among top Sub-Saharan African candidates. Completed Deep Learning and TensorFlow Specializations on Coursera with top marks; earned the TensorFlow Developer Certificate.

## CERTIFICATIONS & AWARDS

- Deep Learning Specialization (Andrew Ng, DeepLearning.AI) — Neural Networks & Deep Learning: 96.5%
- DeepLearning.AI TensorFlow Developer Specialization · TensorFlow Developer Certificate (Google)
- Applied Concurrency in Go · Cyber Resilience (LRN Legal Compliance and Ethics Center)
- Valedictorian, 2020 (Yeshua Educational Institute)

## SELECTED PROJECTS

**Anyscale/Ray Migration (TripAdvisor):** Re-platformed ML pipelines and offline feature store from Spark/Kubeflow to Snowflake + Ray on Anyscale, achieving a ~360x runtime reduction (~4 hours to ~40 seconds).

**Adugbo:** Local-first neighborhood information platform for Nigerian cities (Next.js 14 App Router, Tailwind, NextAuth with Google/Apple OAuth, Go microservices). Geofencing, content seeding, verification, and moderation flows.

**RAG System (ExcelMind):** Retrieval-augmented generation system using NLP models, vector retrieval, and structured output generation for contextual exam explanations.

**ML Survival Prediction (Thesis):** ML models predicting colorectal cancer patient survival from clinical features, covering feature engineering, model selection, and evaluation.

**Weakly-Supervised Pothole Segmentation:** U-Net with ResNet-34 encoder trained with partial cross-entropy loss and focal weighting; experiments on label density and transfer learning.

**Medical Suggestion API (Self-employed):** AI-powered API generating personalized health recommendations, deployed to production.